

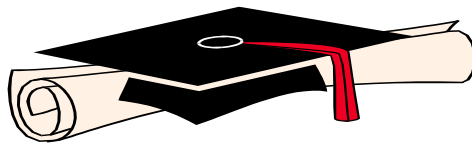


EPC

Elementary Programming with C

Semester: 1
Duration: 90 minutes

The Exam Code: 01
Total Marks: 20



Practical Examination Paper

Do not write on this question paper and return it to the Invigilator after the examination.

Question 1:

[4]

Write a program to ask user input two numbers: number1 and number2. Calculate the sum of all numbers and the total count of odd numbers that fall between the two entered numbers and display the output. (Note: Note that if the first number is greater than the second number, the program will require the user to input the numbers again.)

Sample output:

```
Enter the first number: 5
Enter the second number: 20
The sum of numbers between 5 and 20 is: 135
Total count of odd numbers between 5 and 20 is: 8
```

Question 2:

[4]

Write a program that asks the user to enter 5 decimal numbers, for example: 5, 6, 4, 5, 3, and stores all the values in an array. The program should then reverse this array.

Sample output:

```
Input 5 numbers:
number 1: 5
number 2: 6
number 3: 4
number 4: 5
number 5: 3
Reverse list number you inputted: 3,5,4,5,6
```

Question 3:

[12]

Create a student management system program in C

Note that you must use the pointer concept to do this task.

Create a data structure following the specifications

[1]

Data type	Field	Description
char	name[20]	Name of student
char	country[20]	Country of student
int	birthyear	Birth year of student
float	mark	mark of student

Create menu based program as follows:

[2]

Student management system
1. Input students 2. Display Students 3. Save file 4. Open File 5. exit

Input list students:

[2]

When the user selects the option “1. Input students” from the menu

do the followings:

- Ask the user to enter the total number of Students
- Input information for each Student in the list
- Validate data input by the following rules:

Data anomalies	Message
$18 \geq \text{current_year} - \text{birthyear} \geq 6$	student age must less than 18 years old and greater than 6 years old
$0 \leq \text{mark} \leq 10$	Student mark must less than 10 and greater than 0

- If data input is invalid, ask the user to re-enter.

Display students: [2]

When the user selects the option “2. Display Students” from the menu

- Sort students by name in ascending order.
- Displays details of each Student in the list after the above re-arrangement.

Save to file: [2]

- When the user selects the option “3. Save file” from the menu, do the followings:
- Ask the user to enter a file name.
- Create the file in binary mode.
- If the program cannot open a file, notify the user about that error and return to end this function.
- Write data from the list into the file.
- Notify the user that data has been saved successfully.
- Close the file.

Open file: [2]

When the user selects the option “4. Open file” from the menu, do the followings:

- Ask the user to enter a file name.
- Open the file in binary mode.
- If the program cannot open a file, notify the user about that error and return to end this function.
- Read data out of the file, fill into the list and display them all.

- Notify the user that data has been saved successfully.
- Close the file

Exit:

[1]

When the user selects the option “5. Exit” then exit the program.